FATHIMATHUL MINA CH

AI ENGINEER

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CONTACT

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EDUCATION

2023-Present
GOVT. COLLEGE THALASSERY
- Msc Computer Science with Specialization in AI

2020-2023
NAM COLLEGE KALLIKANDY
- Bsc Computer Science
- 74%

SKILLS

Programming Languages:
- C, C++, Python, Java

Web Development:
- HTML, JavaScript, Node.js, Express

Artificial Intelligence & Machine Learning:
- Deep Learning (CNNs, RNNs, Transformers)
- Computer Vision (OpenCV, Dlib, Face Detection)
- Natural Language Processing (NLP, Sentiment Analysis)
- Model Fine-Tuning (Wav2Vec2, BERT, ResNet)
- Generative AI (Gemini API, LLM)

PROFILE

AI-focused Computer Science student with expertise in ML/DL (Transformers, CV, NLP) and full-stack development (Flutter, Node.js, Python). Experienced in building end-to-end AI systems and optimizing models for deployment. Seeking to apply technical skills in professional AI engineering.

PROJECT

Speech Emotion Detection Application | Final Year Project GitHub Link
- **Technologies: Python, Transformer Models (Wav2Vec2), Flutter, Node.js, Express, MongoDB**
- **Duration: Feb/2025 - Present**
  - Built a real-time emotion detection system by fine-tuning Wav2Vec2, achieving 85% accuracy (TESS)
  - Integrated Gemini API to generate personalized wellness tips (e.g., "Try meditation" for sadness) based on detected emotions.
  - Developed Flutter app with auth and dynamic emotion trend visualization.
  - Creating Node.js/Express backend to process audio, run ML predictions, and store data in MongoDB.

Online Student Engagement Detection GitHub Link
- **Tech: Python, TensorFlow/Keras, OpenFace, CNN, ResNet, Google Colab, SMOTE, DAiSEE**
  - Developed a deep learning-based system to detect student engagement levels in online learning environments using video analysis.
  - Trained a hybrid CNN-ResNet model on the DAiSEE dataset after balancing it with SMOTE and data augmentation, improving accuracy by 60%
  - Utilized OpenFace for facial feature extraction (e.g., gaze direction, head pose, facial landmarks) to analyze engagement cues like attention and focus.

Machine Learning and Deep Learning Pipelines GitHub Link
- Built end-to-end pipelines for data preprocessing, model training, and evaluation.
- Utilized Python with TensorFlow, Keras, and scikit-learn for development.
- Optimized models with hyperparameter tuning and deployed for predictions.